



HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

Instructor: Maria Samad

Date: February 24th, 2025

Quiz # 2

Time Allowed: 45 minutes

Max Points: 30

Name: _____

QUESTION 1: ALGORITHM DESIGN USING BINARY SEARCH (10 marks)

- **Finding the middle location of the given target value in an array of numbers using BINARY SEARCH:**
 - A fixed sized array is given that contains n integers, where 'n' > 1, while the rest of the elements in array are None, if needed. Assume both n and array are in the correct format, so you may use them directly WITHOUT taking any inputs. However, they must be declared/defined in the algorithm properly.
 - The user inputs a target value that is a number, which needs to be searched in the array
 - Design an algorithm that uses binary search to find the middle index of the target value.
 - For example:
 - If given array is [2, 3, 3, 3, 7, 7, 9, None], and given target is 3, then the middle occurrence index of 3 will be the one on index 2 in the array, so the algorithm should return 2
 - If target is not found, then it should just return -1
 - Similarly, if the given target is 7 for the above-mentioned array, then middle occurrence index of 7 is the floor value of middle, and it will return 4
- **REQUIREMENTS:**
 - Do not **HARD CODE** your design for the given example. The algorithm should work for any valid sized arrays.
 - Declare all the data structures and variables being used in this algorithm, including any inputs, if any.
 - **DO NOT JUST SAY "PERFORM BINARY SEARCH TO DO ...", ETC. INSTEAD SPECIFY THE ACTUAL LOGIC FOR IT ON HOW IT IS BEING IMPLEMENTED HERE!**
 - You **MUST NOT** use any additional data structure (including lists/arrays) apart from the given input array.
 - You **MUST NOT** modify the array to forcefully convert it to a Python list just to perform your binary search algorithm
 - You **MUST USE BINARY SEARCH, OTHERWISE, NO MARKS WILL BE AWARDED IF ANY OTHER SEARCH IS USED.**

QUESTION 2: PRIORITY QUEUE OPERATIONS (10 marks)

A priority queue can be implemented using fixed-sized arrays or lists. Assume the given priority queue is implemented using fixed-sized arrays. The largest number represents the highest priority for the given Priority Queues. For a given Priority Queue, called PQ with size = 4 and empty slots represented as -1, complete the following table:

OPERATION	RETURN VALUE	UNSORTED PQ CONTENTS	TIME COMPLEXITY	SORTED PQ CONTENTS	TIME COMPLEXITY
PQ.is_full()					
PQ.enqueue(37, "D")					
PQ.enqueue(17, "S")					
PQ.is_empty()					
PQ.enqueue(57, "A")					
PQ.dequeue()					
PQ.dequeue()					
PQ.enqueue(27, "I")					
PQ.peek()					
PQ.enqueue(20, "S")					
PQ.enqueue(35, "F")					
PQ.dequeue()					
PQ.peek()					
PQ.enqueue(2, "U")					
PQ.enqueue(41, "N")					

QUESTION 3: SELECTION SORT (10 marks)

Sort the given list in descending order using selection sort algorithm.

- [33, 12, 41, 94, 27, 65, 79, 16, 58, 90, 2]

SHOW ALL INTERIM CALCULATIONS IN ORDER TO GET COMPLETE/FULL MARKS. An answer without the intermediate steps will directly result in 0 marks.

